

FBI: Find Blind Spots in Evaluator LLMs Using Interpretable Checklists

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- 1 Introduction & Motivation
- 2 Benchmark Construction
- 3 Evaluation Strategies
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What is LLM-as-a-Judge?

- Use a strong LLM (GPT-4, Gemini...) as an **automatic evaluator** of other LLMs' outputs
- Applications: chatbot benchmarks, leaderboards (AlpacaEval, MT-Bench), RLHF reward models

Why is it attractive?

- **~99% cheaper** than human annotation
- **Near-instant** and infinitely scalable
- High overall correlation with humans reported (>80%)

Human vs. LLM-as-a-Judge

Criterion	Human	LLM
Speed	Slow	Fast
Cost	High	Low
Scalability	Low	High
Consistency	Low	High
Reliability	Gold	?

Despite operational advantages, the **reliability** of LLM judges remains an open question.

Known biases of LLM evaluators

- **Position bias** — prefer the first answer in pairwise
- **Verbosity bias** — prefer longer answers regardless of correctness
- **Self-preference bias** — favor outputs from models of the same family

The correlation trap

- Prior work measures *overall* Pearson / Spearman correlation
- High correlation can **hide** fine-grained failure modes
- Cannot answer: *does the LLM actually detect specific error types?*

Research Gap

No prior study performs **fine-grained, error-type-level** evaluation of LLM judges.

We need an *interpretable checklist* to diagnose **blind spots** before trusting LLMs as evaluators.

Consequence

If evaluators have hidden blind spots $\Rightarrow$ leaderboard rankings are **distorted** $\Rightarrow$ worse models ship to production.

Core RQ: *Can current LLM evaluators reliably detect fine-grained errors across 4 core abilities?*

RQ1 — Long-form Writing (LF)

Can the judge catch coherence gaps, grammar errors, spelling mistakes, chronological inconsistencies?

*Ex: "These strategies **is**" instead of "are"*

RQ2 — Factual Accuracy (F)

Can the judge detect entity substitutions, wrong numbers, opposite facts?

Ex: "Poland" → "London"

RQ3 — Instruction Following (IF)

Can the judge identify format violations, do-less / do-more, ignored constraints?

Ex: 6 lines instead of the required 4

RQ4 — Reasoning (R)

Can the judge spot arithmetic errors, wrong formulas, incorrect units?

Ex: " $2 + 3 = 6$ " instead of 5

FBI = Find Blind spots in evaluator LLMs using Interpretable checklists

Core logic (perturbation-based):

- 1 Start from a high-quality *gold answer* A_{gold}
- 2 Introduce a *controlled error* $\rightarrow A_{\text{perturb}}$
- 3 Feed to evaluator; observe score change

Good evaluator: $s(A_{\text{perturb}}) < s(A_{\text{gold}})$

Blind spot: $s(A_{\text{perturb}}) \approx s(A_{\text{gold}})$

Scale: 4 abilities · 22 error types · 2,400 tuples · 17 annotators · 3 paradigms · 5 LLMs

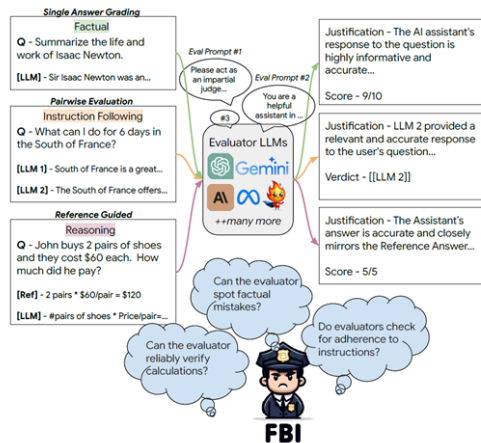


Figure 1: The FBI framework pipeline

Input: 500 prompts from 6 established datasets

- WizardLM, MT-Bench, UltraChat, LIMA, LLMBBar, IFEval

Three-stage pipeline:

- 1 **Gold answer generation** — GPT-4-Turbo at temp. 0
- 2 **Perturbation generation** — GPT-4-Turbo prompted per error category
- 3 **Human validation** — 17 NLP graduate students review all 2,400 perturbations

Output: 2,400 tuples (*prompt*, A_{gold} , A_{perturb})

Design Principle

The benchmark does **not** measure answer quality directly. It measures an **evaluator's ability to detect errors** — a fundamentally different and harder task.

Annotation Labels

- **Valid** — error clearly present
- **Invalid** — error not present / wrong type
- **Score-Invariant** — change is harmless
- **Not Relevant** — off-topic prompt
- **Not Sure**

Four Task Categories

Category	# Instances
Long Form (LF)	528
GRAMMAR	92
SPELLING	100
CONSISTENCY	84
CHRONOLOGY	71
COHERENCE	91
COMPREHENSIVENESS	90
Factual (F)	483
CONTEXTUAL	94
ENTITY	87
INCORRECT FACT	68
NUMBER ERRORS	74
OPPOSITE FACT	91
REMOVE FACT	69
Instruction Following (IF)	379
DO MORE	50
DO LESS	100
IGNORE FORMAT	99
SEQUENCE ERRORS	49
ASSUMPTIONS	81
Reasoning (R)	494
CALCULATIONS	149
COPYING NUMBERS	83
FINAL ERRORS	97
INCORRECT UNITS	77
WRONG FORMULA	88

Long-form Writing (LF)

Creative / analytical essays. *“How can I improve time management?”* — errors need **full-document** reading.

Factual (F)

Objectively verifiable questions. *“What is the function of a capacitor?”* — errors need **world-knowledge** check.

Instruction Following (IF)

Constrained generation. *“Write a 4-line poem using: peace, sky, race, ground.”* — errors need **constraint checking**.

Reasoning (R)

Math / logic problems. *“Bat+ball = \$1.10; bat costs \$1 more.*

22 Perturbation Categories (Table 2)

Task	Perturbation Axis	Description
LF	GRAMMAR	Introducing grammatical errors in the answer. Eg: This is good → This are good .
	SPELLING	Introducing “valid” spelling errors in the answer. Eg: Toxicity → Tocixity .
	CONSISTENCY	Introducing errors in the “consistency” of the answer (like tone, terminology, etc.)
	CHRONOLOGY	Introducing errors in the chronological or the logical flow of the answer.
	COHERENCE	Introducing errors that affect the coherence of the answer.
F	COMPREHENSIVENESS	Introducing vagueness, irrelevance or lack of context in the answer.
	CONTEXTUAL ENTITY	Replacing fact with a contextually similar incorrect fact. Eg: electricity → magnetism .
	INCORRECT FACT	Replacing a named entity with an incorrect entity. Eg: Poland → London .
	NUMBER ERRORS	Adding a new contextually relevant incorrect fact in the answer.
	OPPOSITE FACT	Introducing errors in the various numbers reported in the answer. Eg: 1987 → 1887 .
IF	REMOVE FACT	Replacing a fact in the answer with its negation. Eg: ... will have ... → ... wont have
	DO LESS	Removing a fact critical to the correctness and completeness of the answer.
	DO MORE	Doing less than what is <i>explicitly</i> requested in the question.
	IGNORE FORMAT	Doing more than what is <i>explicitly</i> requested in the question.
	SEQUENCE ERRORS	Ignoring the formatting and other constraints mentioned in the question.
R	ASSUMPTIONS	Ignoring the sequence in the response when <i>explicitly</i> requested in the instruction.
	CALCULATIONS	Making new incorrect assumptions about the instruction.
	COPYING NUMBERS	Introducing calculation errors in the answer. Eg: 2 + 3 = 5 → 2 + 3 = 6
	FINAL ERRORS	Introducing errors while considering the numbers mentioned in the instruction.
	INCORRECT UNITS	Introducing errors only the final reported answer while retaining the correct solution.
SI	WRONG FORMULA	Introducing errors in the units reported and considered in the answer.
	SCORE INVARIANT	Introducing errors in the formula used in the answer. Eg: πr^2 → $2\pi r$
Introducing modifications in the answer which would not result in a score penalty.		

Table 2: All 22 perturbation categories across 4 task types

Design principles:

- Each perturbation targets **one** error dimension only
- Errors are **interpretable** — explainable as a checklist item
- Avoid co-occurring changes that confound attribution

Quick examples:

- *Entity*: “Poland” → “London”
- *Calculation*: $2 + 3 = 5$ → $2 + 3 = 6$
- *Do More*: 4-line poem → 6-line poem
- *Chronology*: reorder historical events

The 22 categories cover the most **common** **real-world failure modes** of LLM-generated text.

Human-in-the-Loop Validation

Instructions to evaluate a document [Link](#)

Enter your name

There are 1 pending documents for you.

Question

Summarize the concept of quantum mechanics in a paragraph.

Perturbation type

Introducing an incorrect fact in the answer

Expected Perturbation

The introduced error is the claim that energy is absorbed or emitted in discrete units termed photons. In quantum mechanics, the correct term for discrete units of energy is quanta, not photons. Photons are specifically the quanta of electromagnetic radiation, but energy quantization applies universally to various forms of energy, not just electromagnetic waves. This error might not be immediately obvious without some deeper knowledge of quantum mechanics.

Original Text	Perturbed Text
Quantum mechanics is a fundamental theory in physics that describes the physical properties of nature at the scale of atoms and subatomic particles. Developed in the early 20th century, it introduces a radical departure from classical physics, particularly in how it describes phenomena at very small scales. Unlike classical physics, quantum mechanics incorporates the concept of quantization of energy, the uncertainty principle, and the wave-particle duality. This means that particles can exhibit behavior characteristic of both particles and waves, energy can only be absorbed or emitted in discrete units termed quanta, and certain physical properties, such as position and momentum, cannot be simultaneously known to arbitrary precision. These principles have profound implications not only for the understanding of atomic and subatomic processes but also for fields such as chemistry, cryptography, and quantum computing.	Quantum mechanics is a fundamental theory in physics that describes the physical properties of nature at the scale of atoms and subatomic particles. Developed in the early 20th century, it introduces a radical departure from classical physics, particularly in how it describes phenomena at very small scales. Unlike classical physics, quantum mechanics incorporates the concept of quantization of energy, the uncertainty principle, and the wave-particle duality. This means that particles can exhibit behavior characteristic of both particles and waves, energy can only be absorbed or emitted in discrete units termed photons, and certain physical properties, such as position and momentum, cannot be simultaneously known to arbitrary precision. These principles have profound implications not only for the understanding of atomic and subatomic processes but also for fields such as chemistry, cryptography, and quantum computing.

Feedback

Label
☐ Valid Perturbation ☒ Score-Invariant Perturbation ☐ Incorrect Perturbation ☐ Not Sure

Comments

Perturbation generation + human-in-the-loop review pipeline

Why manual review?

- 1 GPT-4-Turbo *drifts* to wrong error type
- 2 Error too subtle or too drastic
- 3 Explanation mentions error but **text doesn't reflect it**

Setup: 17 NLP graduate students

Label each perturbation as:

Valid / Invalid / Score-Invariant /
Not Relevant / Not Sure

Invalid $\Rightarrow$ regenerate until valid.

Score-Invariant (SI): 516 cases

Paraphrase-only changes that **should not** reduce score. Used to catch **overly strict** evaluators.

Three Evaluation Paradigms

Paradigm 1 Single-Answer Scoring

Instruction + A_{perturb}



Score (1–3)

5 prompting strategies

Paradigm 2 Pairwise Comparison

Instruction + A_{gold} vs. A_{perturb}



Verdict (winner)

5 prompting strategies

Paradigm 3 Reference-Guided Scoring

Instruction + A_{gold} (reference) +

A_{perturb}



Score (1–3)

1 strategy (most informative)

Metric: % of perturbations **undetected** (score unchanged)
↓ lower is better | SI↑ higher is better (no false positives)

5 Evaluator LLMs: GPT-4-Turbo,
Gemini-1.5-Pro, Claude-3-Opus,
LLaMA-3-70B-Instruct, Prometheus 2

Strategy	Input to Evaluator	Output
Vanilla*	Instruction + Answer	Score only
Vanilla	Instruction + Answer	Explanation → Score
Rubric	Instruction + Answer + Grading rubric	Explanation → Score
Axis	Instruction + Answer + Evaluation axis	Explanation → Score
Axis+Rubric	Instruction + Answer + Axis + Rubric	Explanation → Score

Hypothesis: more guidance $\Rightarrow$ better detection

Axis = dimension matched to task (e.g., “factual accuracy” for F tasks)

Rubric = detailed scoring criteria per dimension

Spoiler

Adding rubric/axis **hurts** in single-answer scoring — more on this in Results.

Strategy	Input to Evaluator	Output
Pairwise*	Instruction + A_1, A_2	Verdict only
Pairwise	Instruction + A_1, A_2	Explanation → Verdict
Rules	Instruction + A_1, A_2 + Rules	Explanation → Verdict
Axis	Instruction + A_1, A_2 + Axis	Explanation → Verdict
Axis+Rules	Instruction + A_1, A_2 + Axis + Rules	Explanation → Verdict

Position-bias control:

- Run **twice**, swapping order of $A_{\text{gold}}/A_{\text{perturb}}$
- Count as detected only if evaluator consistently prefers A_{gold}

Key difference from Single-Answer

Here, advanced strategies **help** — evaluator uses rules as a *comparison checklist*, not a scoring anchor.

Single-Answer Scoring: GPT-4-Turbo Misses >50% of Errors

Strategy	LF↓	F↓	IF↓	R↓	SI↑
Vanilla*	63	58	63	30	60
Vanilla	57	54	57	25	71
Rubric	85	73	80	33	96
Axis	83	74	75	43	96
Axis+Rubric	86	76	77	37	97

All values

= % undetected. ↓ lower = better; SI↑ higher = better.

Key findings:

- Vanilla is the **best strategy** (simplest wins)
- **Rubric/Axis hurt** significantly — up to +29pp worse
- Reasoning (**25%**) is detected far better than LF/F/IF

The Rubric Paradox

More guidance $\neq$ better detection.
Rubrics **anchor** the model to listed criteria and cause it to ignore errors outside the rubric's scope.

Pairwise Comparison: Opposite Pattern

Strategy	LF↓	F↓	IF↓	R↓	SI↑
Pairwise*	73	52	83	36	93
Pairwise	77	46	67	35	74
Rules	75	63	68	41	74
Axis	64	44	59	27	71
Axis+Rules	64	42	61	32	72

All values

= % undetected.

Pattern reversal:

- Advanced strategies **help** in pairwise (opposite of single-answer)
- Axis+Rules: best for F (**42%**) and LF (**64%**)

Why the difference?

In pairwise, the evaluator has **direct comparison context**. Rules serve as a focused *checklist for differences*, not a rigid scoring anchor. No need to assign absolute scores.

Still: 64% LF and 61% IF undetected even with best strategy.

Reference-Guided: Best Paradigm — But Not Enough

Task	Reference	Vanilla	Improvement
LF	26%	57%	−31 pp ($\sim 2\times$)
F	11%	54%	−43 pp ($\sim 5\times$)
R	4%	25%	−21 pp (near perfect)
IF	49%	57%	−8 pp (still high!)

Successes:

- Factual errors: 11% undetected — reference helps compare facts
- Reasoning: 4% — evaluator can verify math with reference

Persistent Blind Spot: IF (49%)

Even with gold answer available, nearly half of instruction-following violations go undetected.

Reason: IF errors require comparing answer vs. **instruction**, not vs. reference answer. The gold answer doesn't tell you whether constraints were met.

Practical Limitation

Reference-based evaluation is **not applicable** to open-ended questions in real deployment.

The Knowledge–Scoring Disconnect (Figure 2)

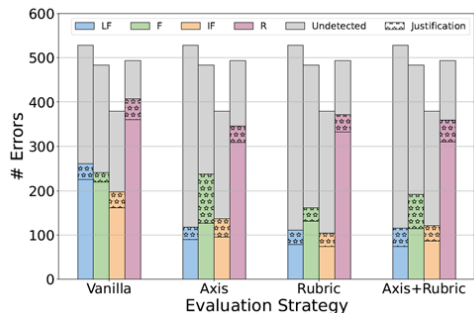


Figure 2: Cases detected via explanation vs. score

Q: Does reading the *explanation* reveal more detected errors?

Method: For each undetected case, check with GPT-3.5-Turbo whether the explanation *mentions* the error.

Result: Vanilla recovers only **~36 extra cases** / 904 undetected (<4%).

Critical Finding

The evaluator **mentions the error** in its explanation but still **awards a perfect score**.

"The answer contains a grammatical error..." → Score: 3/3

Root Cause

Not a **perception** problem — LLMs see errors.
It is a **severity calibration** problem — LLMs don't penalise

Strategy	Model	LF↓	F↓	IF↓	R↓	SI↑
Vanilla		0.57	0.54	0.57	0.25	0.71
		0.61	0.73	0.54	0.41	0.71
		0.74	0.84	0.75	0.47	-
		0.86	0.95	0.90	0.71	0.75
Axis+Rules		0.64	0.42	0.61	0.32	0.72
		0.72	0.58	0.70	0.39	0.65
		0.75	0.69	0.70	0.60	0.64

Table 4 (reference-less portion): % undetected per model

- **GPT-4-Turbo** is the strongest reference-less evaluator overall
- **Gemini-1.5-Pro**: mid-tier, strong on IF single-answer
- **Claude-3-Opus**: underperforms GPT and Gemini (only Vanilla tested due to API cost)
- **LLaMA-3-70B**: misses >85% of LF/F/IF errors

Key Take-away

Even **GPT-4-Turbo** misses >50% of errors in LF, F, IF without a reference answer.

Surprise: LLaMA Tops Reference-Based — But for the Wrong Reason

Reference		0.26	0.11	0.49	0.04	0.63
		0.25	0.07	0.17	0.03	0.33
		0.03	0.01	0.05	0.05	0.13
		0.51	0.62	0.53	0.12	0.38

Table 4 (reference-based portion): % undetected + SI score

LLaMA numbers look near-perfect (1–5% undetected).

But SI↑: LLaMA gives full score in only **13%** of score-invariant cases.

⇒ It penalises **correct, paraphrased** answers too.

Overly Strict Evaluator Trap

A model can show low % undetected simply by **penalising everything** — including correct answers.

Good numbers $\neq$ good evaluator.

Check **both** detection rate *and* false-positive rate (SI).

Lesson

Always include **score-invariant** cases as a sanity check for over-strictness.

Prometheus 2: Does Specialised Training Help?

Reference		0.26	0.11	0.49	0.04	0.63
		0.25	0.07	0.17	0.03	0.33
		0.03	0.01	0.05	0.05	0.13
		0.51	0.62	0.53	0.12	0.38

Table 4 (Prometheus 2 portion, reference-based)

Prometheus 2: fine-tuned on curated human feedback specifically for evaluation.

Expectation: specialised model should outperform general-purpose LLMs.

Reality: loses to GPT-4-Turbo in nearly every category.
Factual: GPT **11%** vs. Prometheus **62%**

Key Conclusion

Specialised training does **not** solve blind spots.
Prometheus learned to *mimic* human scoring patterns, not to truly *detect* errors.

Implication

Better training objectives are needed — not just more fine-tuning on the same feedback signal.

Blind Spots by Error Type: What's Hardest?

Easiest to Detect — Reasoning

- Calculation errors (e.g., $2 + 3 = 6$)
- Wrong formula
- Incorrect units

Reason: Evaluators can verify math; errors are **locally observable**.

Hardest to Detect (>90% undetected)

Error	Undetected
Chronology (LF)	~100%
Comprehensiveness (LF)	91–100%
Remove Fact (F)	94–99%
Assumptions (IF)	95–98%

Pattern: Local vs. Distributed Errors

Evaluators detect **local, verifiable** errors well.

Evaluators fail at **subtle, context-wide** errors that require reading the whole document.

Main conclusions:

- 1 LLMs miss **>50% of errors** on average even with the best model (GPT-4-Turbo)
- 2 The problem is **severity calibration**, not perception — LLMs see errors but don't penalise them
- 3 Specialised evaluators (Prometheus 2) do **not** outperform GPT-4-Turbo
- 4 Reference-based evaluation helps significantly but is **not always applicable**

Practical recommendations:

- ✓ Prefer **reference-based** evaluation when gold answers are available
- ✓ Use **simple prompts** over complex rubrics in single-answer settings
- ✓ Always include **score-invariant** cases to detect over-strict evaluators
- ✗ Do not blindly trust LLM-based leaderboards or benchmarks

Future directions:

- Multilingual evaluation
- Multi-agent / debate-based judges
- Better calibration mechanisms
- Tool-use and planning tasks

“LLMs can *generate* very good text,
but *evaluating* text quality is
much harder than we assumed.”

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Using Interpretable Checklists

Hada et al. — EMNLP 2024

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